Homework Assignment
Chapter 1: Foundation of Engineering Economy

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Section 1: Cash Flows

Case Study: A manufacturing company is evaluating a *4-year* project to upgrade its assembly line. The engineering team has gathered the following financial *estimates* to conduct an economic analysis:

- **Bank Loan:** A $100,000 loan is received from an industrial bank today to help fund the project.
- **Equipment & Installation:** The equipment cost is $105,000. However, the installation cost depends on local labor rates and could be anywhere between $10,000 and $15,000. For the baseline analysis, the team assumes exactly $15,000 for installation (the worst case scenario).
- **Annual Savings:** The new line will generate $35,000 in labor and material savings each year.
- **Annual Maintenance:** A yearly maintenance contract will cost $4,000.
- **System Overhaul:** A major system overhaul is required at the end of Year 2, costing an additional $12,000.
- **Salvage Value:** The equipment is expected to be sold for $25,000 at the end of its life in Year 4.

Exercise 1.1: Identifying Cash Inflows and Outflows
From the perspective of the company, classify each of the six items in the case study as either a **cash inflow (+)** or a **cash outflow (-)**.

Exercise 1.2: Point vs. Range Estimates
Based on the project description, identify which equipment-related cost represents a **point estimate** and which represents a **range estimate**. Briefly explain the advantage and disadvantage of each type of estimate.

Exercise 1.3: Net Cash Flow Calculation (Initial Investment)
Calculate the **net cash flow** at Year 0 (the initial investment phase).

Exercise 1.4: Net Cash Flow Calculation (Operations)
Calculate the **net cash flow** at the end of Year 1, Year 2, Year 3, and Year 4.

Exercise 1.5: Drawing a Cash Flow Diagram
Using the net cash flows calculated in Exercises 1.3 and 1.4, draw the complete **cash flow diagram** for this 4-year project from the company's perspective. Ensure the timeline is properly labeled and arrows correctly indicated.

Exercise 1.6: Cash Flow Diagram - Software Upgrade
An IT department is implementing a new cloud infrastructure. The financial projections are as follows:

- An initial investment of $50,000 is required today for software licenses and training.
- Savings from reduced server maintenance will start at $15,000 in Year 1 and will increase to $20,000 in Year 2, $25,000 in Year 3, and $30,000 in Year 4.
- In Year 3, the company must pay a one-time licensing renewal fee of $8,000.
- At the end of Year 4, the software will be retired, and the company will incur a $5,000 decommissioning cost.

Determine the **net cash flow** for each year and draw the corresponding **cash flow diagram** from the IT department's perspective.

Exercise 1.7: Cash Flow Profile of a Construction Project
A civil engineering firm signs a contract to build a small bridge. The project has a 6-month timeline, and financial transactions are recorded monthly:

- **Month 0:** Initial mobilization cost of $40,000.
- **Months 1, 2, and 3:** Material and labor costs of $25,000 per month.
- **Month 3:** The firm receives a progress payment of $60,000 from the client.
- **Months 4 and 5:** Finishing labor costs of $10,000 per month.
- **Month 6:** Final inspection and handover cost of $5,000, along with the final payment of $80,000 received from the client.

Create a table showing the **monthly net cash flows** for Month 0 through Month 6, and construct the **cash flow diagram**.

Section 2: Simple and Compound Interest Calculations

Exercise 2.1: Simple vs. Compound Interest
An engineer needs to borrow $20,000 to start a small contracting business and plans to repay the entire amount (principal plus all accumulated interest) in a single lump sum at the end of 5 years.

- **Bank A** offers a simple interest loan at 8.5% per year.
- **Bank B** offers a compound interest loan at 7.0% per year.

Calculate the total amount owed at the end of 5 years for both options. Which bank offers the more economical deal for the engineer?

Exercise 2.2: Rate of Return and MARR

An investor purchases shares in a renewable energy startup for $50,000. Exactly one year later, the investor sells the shares for $54,500.

a) Calculate the interest amount and the **rate of return** on this investment.

b) If the investor's Minimum Attractive Rate of Return (MARR) is 10% per year, was this a mathematically sound investment? Explain briefly.

Exercise 2.3: Economic Equivalence

A company has the option to pay a supplier $50,000 today or $54,000 exactly one year from now. If the company's Minimum Attractive Rate of Return (MARR) is 10% per year, which payment option is more economical? Support your answer using a **Present Worth** analysis (i.e., by finding the equivalent value of next year's payment in today's dollars).

Exercise 2.4: Compound Interest over Multiple Periods

A civil engineer deposits $10,000 into a high-yield corporate savings account that earns 6% compound interest per year.

a) How much money will be in the account at the end of 4 years?

b) What is the total interest earned over this 4-year period?

Exercise 2.5: The Rule of 72

A manufacturing plant is considering an automation project that promises a 9% annual compound rate of return.

a) Using the **Rule of 72**, estimate how many years it will take for the invested capital to exactly double in value.

b) If the company's management requires all new projects to double their value within 6 years, does this project pass the initial screening?

Exercise 2.6: Deriving the Rule of 72 ☕

a) Derive the Rule of 72. *Hint: Set the future value to twice the principal.*

b) After deriving the rule, show why the rule works best when the interest rate is between 5% and 12%.

Exercise 2.7: The Rule for Simple Interest ☕

The Rule of 72 does not work for simple interest. Let's find the equivalent rule for simple interest.

a) Derive the rule.

b) Based on your result, state the "Rule" for simple interest. If a bank offers an 8% annual simple interest rate, how long will it take to double your investment? Compare this to the Rule of 72 estimate for 8% compound interest.

Section 3: Introduction to Spreadsheet Functions

Exercise 3.1: Describing Financial Functions

Spreadsheets are essential tools for analyzing engineering economy problems. Match the following standard spreadsheet financial functions to their correct description.

(Functions: `FV`, `PV`, `PMT`, `RATE`, `NPER`)

1. Calculates the present value of a loan or an investment based on a constant interest rate.
2. Calculates the future value of an investment based on a constant interest rate and payments.
3. Calculates the payment amount for a loan based on constant payments and a constant interest rate.
4. Calculates the number of periods (e.g., years or months) required for an investment or loan.
5. Calculates the interest rate per period of an annuity or loan.